

ADHAM EHAB ERFAN, B.Sc.

Communication & Electronics Engineer | Wireless, RF, Embedded Systems & Automation
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PROFILE

Communication and Electronics Engineer and Teaching Assistant pursuing an M.Sc. with multidisciplinary academic experience across wireless communication, embedded systems, signal and image processing, and advanced telecommunication systems. Specializing in the intersection of wireless networks and deep learning. With hands-on research experience in MIMO systems and neural codebook design, I am seeking a Master's/PhD scholarship to drive advanced, research-intensive projects in wireless communication, 5G/6G signal processing, deep learning and communication theory, or any other related fields.

CORE EXPERTISE

- MIMO & Antenna Systems • 5G NR Physical Layer (SSB/CSI-RS) • Channel Estimation & MMSE Equalization
- Beamforming & Codebook Design • Spectrum & Propagation Analysis • Multi-path Channel Modelling
- Interference Management • Ray-Tracing Simulation (Sionna) • MATLAB RF / Comms Toolboxes
- Embedded RF (ESP32, ARM) • Technical Documentation

EDUCATION

M.Sc. in Communications & Electronics Engineering | Alexandria University | Egypt 2026 – Present
GPA: 4.00/4.00

Thesis: Optimization of Direction of Arrival “DoA” using optimization techniques HHO and PSO.

B.Sc. in Communications & Electronics Engineering | Alexandria University | Egypt 2021 – 2026
GPA: 3.57/4.00

Graduation project (Excellent): Neural Codebook Design for MIMO Beam Management by building an end-to-end deep learning pipeline for SSB and CSI-RS codebook generation in multi-cell 5G networks, simulated using the Sionna ray-tracing library.

Thanawya Ama | Omar Ibn El Hattab | Egypt 2020 – 2021
Grade: Excellent 3rd in school
Core Focus: Advanced Mathematics, Mechanics, and Physics

EXPERIENCE

Teaching Assistant 2026 – Present
AASTMT | on-site

- Teaching communication and electronics courses: signal and systems, embedded systems, digital communication, image processing, and communication systems.
- Supervising weekly laboratory sessions for courses including Electronic Circuits, Digital Signal Processing (DSP), and Communication Systems.
- Teaching software-based simulation tools, guiding students through MATLAB, Simulink, NI Multisim, PSpice, and Proteus exercises.
- Managing the evaluation and testing of final course projects, ensuring designs met specified electrical and performance criteria.

English Instructor 2025 – Present
IEEE SSCS | Hybrid

- Designed and delivered an intensive 8-week Academic IELTS preparation curriculum, specifically structured to help students achieve a target band score of 7.0 or higher.
- Equipping students with advanced test-taking techniques, including time-management strategies for complex reading passages, structural frameworks for essay writing, and fluency tactics for speaking interviews.
- Formulating a continuous evaluation system using targeted quizzes and full-length mock exams to benchmark student progress, identify individual weaknesses, and provide actionable, personalized feedback.

AI Trainer 2025 – Present
RWS | Remote

- Cleaning datasets, doing transcription for audios, and translating EN/AR.

Mathematics Teacher

2023 – 2024

Center | On-site

- Taught high school students (first and second level).

TRAINING & INTERNSHIPS

R&D

2024 – 2026

University Center for Career Development | Alexandria, Egypt

- Contributed to technical documentation, research, and data analysis.

Embedded Systems Intern

Summer 2024

Siemens | Cairo, Egypt

- Developed startup code and peripheral drivers for AVR and ARM Cortex-M microcontrollers; performed memory mapping and low-level hardware bring-up.
- Implemented and debugged UART, SPI, I2C serial protocols for sensor interfacing and inter-device communication in an industrial environment.
- Programmed ESP32 for real-time wireless data acquisition and cloud streaming; practiced hardware–software integration with oscilloscope and logic analyzer.

Networks Training (CCNA)

Summer 2023

ICG | Alexandria, Egypt

- Designed and simulated a multi-subnet LAN in Cisco Packet Tracer utilizing RIP/OSPF protocols to enable dynamic routing between distinct gateways.

TECHNICAL SKILLS

Expert

Advanced

Intermediate

Tools

MATLAB, DSP Toolbox, RF Toolbox, Simulink, MIMO Systems, Beamforming
Python, Signal Processing, SDR, Machine Learning (NumPy, TensorFlow), CNNs, Autoencoders, DQN
C, C++, Embedded Systems (AVR, ARM), UART/SPI, IoT Development
LATEX, Microsoft office, Proteus, Multisim, OrCAD, ROS, Git, AutoCAD

LANGUAGES

English Advanced B2

French Basic A1

CERTIFICATES

• Embedded Systems — ITI	2025
• IELTS Academic — Band 6.0	2024
• Deep Learning — NVIDIA	2024
• Machine Learning — Generative Technology Center	2023
• Embedded Systems Essentials — SIEMENS	2023

EVENTS

- **Job Fair**— Faculty of Engineering, Alexandria University
- **Technical Summit** — Bibliotheca Alexandrina
- **ICT Exhibition** — Cairo (International Communications & Technology)
- **Cairo International Book Fair**— Cairo

PERSONAL PROJECT & APPLICATIONS

Islamic MVP — Mobile Application— Prayer Tracker

- Built a prayer time tracker application with daily scheduling, notifications, and progress logging.

MailWatch — Mobile Application— Gmail & WhatsApp Notifier

- Built an app that scans the user's inbox and spam folder for specified emails from the user and notifies on WhatsApp.

UCCD — Mobile Application— Event and Workshops manager

- Built a full-stack event application enabling administrators to publish workshops, issue digital tickets, and streamline check-ins using a built-in QR reader.